Coffee – The Eye Opener?

Author: Tracey Williams CFNC - September 2026
AI Assisted Cross-Reference Checked: Ezra Darash

It's Personal

Y'all, it's September already! But you would not guess it with temperatures still around 100 degrees outside and no rain in the immediate forecast to cool things down! So, I have been on a month-long hiatus from writing, instead spending my time working on my first version of the Basic Biblical Nutrition (BBN) website—and what a blast it has been to be a code jockey again  —designing and developing webpages, writing code, and testing all the cool stuff on this amazing site... and YES!! Even using AI to create the pages and content layout that I could not have dreamt up and created from scratch!!

Ok…Now, I bet you are wondering how the topic of coffee made it into my **Take 2** cut… Well, I'll tell you. When the BBN webpage was completed, it needed to be dry-run and tested. I asked a few friends to review and comment and I received my first **Take 2** question from a reader asking "*Is Coffee Good or Bad for You?*" followed by:

> *"I drink coffee every morning, 2-3 cups of regular. Should I be using any additives? Drink more? Drink less? Thanks"*

When asked, he added that by additives he meant milk, half-and-half, cream and even butter. So, without further ado, let's find out what makes coffee tick!

A Brief Tour of Coffee History…

(If y'all don't like a little history and don't want to fall asleep on me… pour yourself a nice, tall glass of cold, refreshing Matcha Iced Tea. It will surely get ya' hoppin'!) So, let's begin the coffee saga. The true origins of coffee are somewhat obscured by time and legend, but the coffee plant *Coffea arabica* is native to the highlands of Ethiopia. While stories such as the famous Ethiopian goatherd Kaldi describe how coffee was "discovered," there is no historical evidence confirming the tale. What is better documented is that coffee cultivation and preparation developed in Yemen, where coffee became an important part of religious and social life by the fifteenth century. Sufi scholars used the beverage to help them remain awake during nighttime prayer and study, and coffeehouses soon emerged as gathering places. From Yemen, coffee spread through the Red Sea trade routes to Mecca and Cairo and throughout the broader Islamic world.[2]

By the seventeenth century, coffee had made its way into Europe, where coffeehouses became centers for conversation, commerce, news, and political debate. European colonial powers eventually established coffee cultivation in other tropical regions, including the Caribbean, Southeast Asia, and the Americas, transforming coffee from an expensive luxury into a major global commodity. Brazil later became—and remains—the world's largest coffee producer.[1] Today, coffee is one of the world's most widely consumed beverages, but its long journey from the forests of Ethiopia and the mountains of Yemen to our morning coffee cup is a reminder that this seemingly simple beverage has a remarkably complicated history.

By the way, let's not forget the American relationship with coffee, as it took another interesting turn during the Civil War. By the time of the Civil War, coffee had long since replaced tea as America's preferred caffeinated beverage. *(Oh no! We Southerners love, love, love our tea!! I am boo-hooing in mine.)* For the Union soldiers, *(Those Yankees!)* it became much more than a morning pick-me-up. Coffee was considered an important source of energy and morale—almost as vital as gunpowder. Union General Benjamin Butler reportedly ordered his soldiers to always carry coffee, declaring, "If your men get their coffee early in the morning, you can hold" your position. But, when the Union blockade disrupted coffee imports from Brazil, the shortage became serious enough that finding new sources became a military concern. Then Liberia stepped in, shipping thousands of pounds of coffee to the Union, helping keep the Union soldiers supplied while coffee became increasingly scarce in the Confederacy.[1]

The war may also have helped create America's enduring coffee habit. Soldiers became accustomed to drinking coffee several times a day, and according to Smithsonian historian Jon Grinspan -- many carried those habits home after the war. By 1885, U.S. coffee imports had risen to approximately 11 pounds per person annually—nearly twice the pre-war level.[1] Coffee had moved well beyond an exotic beverage from distant lands; it had become woven into American culture, daily routines, and even military life. And that brings us to the question at hand: **what exactly is in that familiar cup that made coffee such a powerful part of our history—and more importantly, what does it actually do inside the body?**

Did you know? *(Lordy, y'all! We are fixin' to dive right into how coffee and chicory became a time-honored Southern tradition—most notably down in New Orleans. Hold your horses on that chicory, though; we're gonna circle right back to it in just a bit.)* During the Civil War, when the Union blockade disrupted coffee shipments into the city, New Orleanians began adding roasted chicory root to their dwindling coffee supplies to make the coffee last longer. The shortage eventually ended, but the combination stuck. Today, coffee blended with chicory is a signature of New Orleans coffee culture—most famously served at Café du Monde. [18]
[Click here for link to Coffee History Map]

Coffee Chemistry Class 101

(Before we get into what coffee does inside our bodies, we need a little chemistry lesson Sweetie.) Coffee contains several alkaloid compounds, including caffeine, theophylline, and theobromine, with **caffeine being the primary compound responsible for coffee's stimulating effects**. These compounds occur in different amounts, and caffeine is by far the most abundant and physiologically significant of the three in coffee. Caffeine's effects on alertness occurs roughly 30 minutes after drinking, although the timing varies from person to person. [3,4]

Coffee's Impact on the Body

When the caffeine kicks in, it can influence dopamine and adrenaline within the body's central nervous and stress-response systems. Dopamine is a neurotransmitter involved in motivation, reward, attention, and movement. *(Oh, the Thriller chemical—Sing it, Michael!)* [3,4] Caffeine can also stimulate the sympathetic nervous system, contributing to the release of adrenaline, the body's "fight or flight" hormone. As described in Andrea Nakayama's article *The Chemistry of Coffee*: "In an emergency, adrenaline increases your blood pressure, respiratory rate, heart rate, and general level of alertness and anxiety. It also redirects blood from your essential organs to your limbs so you can run or fight. This can disrupt the body's ability to perform normal and parasympathetic functions such as digestion, absorption, detoxification, elimination and reproduction." [3]

However, that is not all that is happening in your body. When you drink a caffeinated beverage such as coffee, you may feel more alert and ready to seize the day—but caffeine is also **blocking adenosine from binding to its receptors in the brain.** Adenosine normally builds throughout the day and contributes to feelings of sleepiness and the body's drive for rest. By blocking adenosine receptors, caffeine temporarily reduces that sleepy signal and helps keep you awake. It is adenosine that is responsible for keeping dopamine in check, as well as, for the energy transfer at the cellular level, modulating internal inflammation, the widening of blood vessels known as coronary vasodilation, and finally, promotes sleep. [3,4]

<u>In essence, coffee stimulates your body and keeps you awake, while adenosine makes you sleepy… Thus, we can feel wired and tired at the same time!!</u>

Pros and Cons of Drinking Coffee

<u>Pros:</u> As we have just read, coffee contains caffeine, polyphenols, and other bioactive compounds that may provide antioxidant and other health benefits. Caffeine can increase alertness, energy, and exercise performance, while research has associated moderate coffee consumption with a lower risk of several chronic conditions, including type 2 diabetes, Parkinson's disease, certain cancers, and cognitive decline. However, as we will soon explore through the 3 Roots of Functional Nutrition—Genes, Digestion, and Inflammation—<u>the potential benefits of coffee may not be experienced the same way by everyone.</u> [3,4,5]

<u>Cons:</u> While coffee can provide benefits, caffeine's stimulating effects can also work against the body's natural need for rest and recovery. Depending on individual sensitivity and caffeine metabolism, coffee may contribute to jitteriness, anxiety, increased heart rate or blood pressure, digestive discomfort, or difficulty sleeping. As we will see through the "3 Roots Many Branches" lens, <u>how a person's body responds to coffee may be just as important as how much coffee they drink.</u> [3,4,5]

A Functional Nutrition Approach

With a Functional Nutrition approach, we look at the chronic health issues through the "3 Roots Many Branches" lens. "Those three roots are the genes (or one's genetic predisposition), digestion, and inflammation." We will look at coffee through these three roots as we analyze the many branches (the signs and symptoms) listed above that can be linked to drinking coffee and of course, caffeine *(that is along for the ride – that free-loader!).*[3] The three Roots as we see them through the Functional Nutrition lens:

Genes: Genetics play a huge role in how the body metabolizes caffeine and how the brain responds. People who metabolize caffeine slowly will be more sensitive to its effects on their body. In fact, some people cannot process caffeine well and even the smallest amount can result in feeling anxious, have the jitters and sleepless nights. As a matter of fact, caffeine metabolism is influenced by two genes: **CYP1A2**, which produces the liver enzyme that breaks down caffeine, and Aryl Hydrocarbon Receptor (**AHR)**, which helps determine how much of that enzyme your body produces. [3] *(Boy that was a mouthful that probably went over your head and under the floorboards! So, to make that statement real Southernly and simple… Just remember there are two genes -- no, not blue jeans! -- that determine how fast you burn through your caffeine. CYP1A2 makes the liver enzyme that breaks the stuff down, and AHR sits on the porch deciding how much of that enzyme your body's gonna brew. Now that makes sense, right?)*

Digestion: Coffee can get the digestive system moving—literally. For some people, coffee stimulates gastrointestinal and colon activity and can produce a bowel movement within minutes. It also stimulates gastric acid and other digestive secretions, which may be helpful for some but may contribute to heartburn, reflux, or digestive discomfort in others. [3,4,6]

Caffeine is also processed primarily by the liver through the CYP1A2 enzyme, as we discussed earlier. That means the liver is certainly involved in processing caffeine, but this does not automatically mean coffee is harmful to the liver. In fact, research has found associations between moderate coffee consumption and better liver health. Once again, the individual's response, amount consumed, and overall health picture matter. [6,9]

Inflammation: Inflammation is another piece of the coffee puzzle, and this is where things can get a little confusing. Coffee contains polyphenols and other compounds that may have antioxidant and potentially anti-inflammatory effects, yet caffeine can also stimulate the body's stress response in some individuals. In other words, coffee may have compounds that support the body's defenses while its caffeine content may produce an unwanted response in someone who is particularly sensitive. Research examining coffee and inflammatory markers has produced mixed results, suggesting that coffee's effects on inflammation are more complicated than simply labeling it anti-inflammatory or pro-inflammatory. [10]

When we are talking about inflammation, we also need to talk about the immune system, because the two are closely connected. And this is where that adrenaline we talked about earlier becomes important. Caffeine can increase the body's stress response, including the release of adrenaline and cortisol. In the short term, this can provide the familiar boost in energy and alertness. But when the body's stress response is repeatedly activated, chronic stress can interfere with normal immune regulation and contribute to inflammation and immune dysfunction. [3,4,11]

This doesn't mean that every cup of coffee causes inflammation or that caffeine causes autoimmune disease. It does mean that the individual's response matters. Someone who is already stressed, exhausted, or particularly sensitive to caffeine may experience a very different response than someone who metabolizes caffeine quickly and drinks coffee without noticeable effects. *(In other words, Dear, sometimes that cup of Joe isn't creating energy—it may be borrowing it from tomorrow's account.)* And once again, the question isn't simply whether coffee is good or bad. It's how **your body** responds to it.

What About Adding Something to Your Coffee?

Remember our reader's original question about milk, half-and-half, cream—and even butter? What you add to your coffee can change the nutritional picture. Milk or half-and-half can add calories and fat, while sweeteners and flavored syrups can add sugar. If you enjoy your coffee black, you are simply avoiding those extras. But if black coffee isn't your style, what you add matters. A little milk, milk substitute, or half-and-half can add creaminess without the sugar found in many flavored creamers. For flavor without added sugar, try a sprinkle of cinnamon, cardamom, cocoa powder, or a touch of vanilla extract. [5]

(Waiter… Don't forget the butter!) Ahhhh! Butter coffee… it has become popular as a high-fat, high-calorie morning beverage, but the evidence does not show that adding butter makes coffee healthier. A small study examining coffee with butter and MCT oil found no significant improvements in measured cholesterol, blood sugar, blood pressure, or inflammation markers after 12 weeks of use. Because the study combined butter with MCT oil and had a relatively small number of participants, it cannot establish a health benefit from butter alone. [15] *(Now for those who do not know what MCT is… it is just a fancy name for medium-chain triglycerides. That's a special kind of fat made up of short little fatty acids that your body can soak right up and turn into energy really quick. Think of it like taking the bypass around town instead of getting stuck behind a tractor on an old back road.)*

So, if you like butter in your coffee, think of it as a **dietary choice—not a health requirement.** Whether it makes sense depends on the rest of your diet, your energy needs, and how your body responds to saturated fat.

Chicory Root: A Prebiotic Alternative

If you are looking for a coffee alternative—especially one without caffeine—chicory root is worth knowing about. Chicory (*Cichorium intybus*) has been used as a coffee substitute for generations. The root can be roasted, ground, and brewed much like coffee, producing a dark, rich beverage with a somewhat coffee-like flavor. Andrea Nakayama also points out that chicory is naturally rich in prebiotic compounds, making it an interesting alternative from a Functional Nutrition perspective. [12]

Chicory root is particularly notable for its **inulin**, a type of soluble prebiotic fiber. Unlike many carbohydrates, inulin is not fully digested in the upper digestive tract. It travels to the colon, where gut bacteria ferment it and use it as a food source. Research suggests that chicory-derived inulin can support beneficial changes in the gut microbiome and bowel function. [13,14]

This makes chicory especially interesting when we look at coffee through our Functional Nutrition lens. Coffee may stimulate the digestive tract, while chicory brings something different to the table: **prebiotic fiber that feeds the microbes living in the gut.** It isn't coffee, and it isn't trying to be—but it can provide a warm, roasted beverage ritual without the caffeine.

Did You Know? If you have ever eaten endive, you've already met chicory! Endive and chicory belong to the same plant family and are both forms of Cichorium. So that fancy little salad green sitting on your plate has a root that can end up in your coffee cup. *(Now that's what I call getting more mileage out of a veggie!)* [12]

One word of caution: because chicory root is rich in inulin and other prebiotic fibers, adding a lot at once may cause gas, bloating, or digestive discomfort in some people. Like with all new food introductions, when introducing chicory root into your diet, it is best to start low and go slow—particularly if you have a sensitive digestive system. [12,13]

How to Kick the Caffeine Addiction

(Oh, good for you! Kicking Morning Joe to the Curb! He probably wasn't your type anyway! 😊 *)* If you decide that caffeine is no longer working for you, you don't necessarily have to quit cold turkey. Research has found that gradually reducing caffeine can produce fewer withdrawal symptoms than stopping abruptly, and current evidence supports tapering as a practical way to make the transition easier. [7,8]

Dr. Frank Lipman offers a simple one-week approach using a combination of regular and decaffeinated coffee. His goal is to gradually reduce caffeine while giving the body time to adjust. [6]

The Five Step Bootcamp to Kicking Morning Joe to the Curb:
Step 1 — Day 1: Have your usual amount of coffee.
Step 2 — Days 2–4: Blend your coffee **50% regular and 50% decaf**.
Step 3 — Day 5–6: Change the blend to **25% regular coffee and 75% decaf**.
Step 4 — Day 7: Eliminate coffee, or choose another lower-caffeine beverage if you prefer.[6]
Step 5 — Give your body time to adjust: Lipman's seven-day plan is one practical way to taper caffeine. The research supports gradual reduction, but it does not establish this particular seven-day schedule as the only or best approach. If withdrawal symptoms are bothersome, the taper can be slowed down rather than forcing the process. [6–8]

(So, you may be wondering…What may I feel when I cut back? Well, I say… Don't worry! Be happy! 'Cause I will be praying for you! 🙏 *)* Caffeine withdrawal is a real physiological response and can occur after substantially reducing or stopping regular caffeine intake. Symptoms commonly include **headache, fatigue, sleepiness, decreased alertness, difficulty concentrating, irritability, low mood, nausea, and flu-like symptoms such as muscle aches or stiffness**. Symptoms can begin within **12–24 hours**, often peak within the first couple of days, and generally resolve within several days. [8]

So, if you decide to **kick Morning Joe to the curb**, don't be surprised if your body complains a little before it realizes you're serious. These symptoms are usually temporary, and gradually reducing caffeine may make the transition more comfortable. [7,8]

One Mo' Thought… Does How You Brew Coffee Matter?

(So Dear, before you go throwing that cup of Joe in my face because you feel you cannot give up your morning 'love', at least consider this…) How coffee is prepared can also affect what ends up in your cup. Coffee beans naturally contain oily compounds called **cafestol and kahweol**. These compounds can raise LDL cholesterol, but a paper coffee filter catches most of those oils before they make it into your cup. Unfiltered methods, such as French press, Turkish, or boiled coffee, allow more of them through. So, if you are watching your cholesterol, **a good ol' paper-filtered cup of Joe may be the better choice.** [16,17]

Concluding Remarks

We started this research project trying to answer questions from one of our readers, "**Is coffee good or bad for you?**" We looked at our long history and love for that dark brew, its basic chemistry and what it does to the body, highlighted the pros and cons of drinking coffee, and finally examined coffee through the Functional Nutrition lens of "3 Roots Many Branches" paradigm. We also looked at alternatives to drinking coffee; breaking the relationship with our morning 'love'; and if you must have that Mornin' Cup of Joe, what to do to help your body metabolize it.

Viewing coffee through the Functional Nutrition lens of **Genes, Digestion, and Inflammation**, the answer turns out not to be quite so simple:

- Coffee affects people differently.
- Genetics can influence caffeine metabolism.
- Digestion can respond differently from person to person.
- And caffeine's effects on stress, sleep, and the immune system may matter more for some people than others.
- We also discovered that what you put in the cup—and how you brew it—can change the nutritional picture.

So, if you enjoy your morning cup of Joe, you don't necessarily need to break up with it. But if coffee isn't your thing—or your body isn't particularly fond of it—there are alternatives, including caffeine-free chicory root.

Simply put… Coffee isn't good or bad—the individual matters.

Reference material available upon request.

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